

Use Case: FootSight Analytics

Project Description:

Football Player Performance & Market Analytics

Team:

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1. Industry & Organization Context

Industry: Sports analytics, football (soccer) clubs, scouting agencies, fantasy sports platforms, sports media, and player agents.

Organization profile:

- Name: FootSight Analytics
- Mission: Provide data-driven scouting, performance evaluation, and market-value insights to clubs and agents to make better recruitment and salary decisions.
- Stakeholders: Sporting Director, Head Scout, Head Coach, Data Analyst, Club CFO, Player Agents.

Why this industry? Football clubs increasingly rely on data to reduce transfer risk, optimize lineups, and quantify player value. The domain is rich in measurable KPIs and business decisions that translate directly to revenue (transfer profits, competition results, sponsorships).

2. Business Problem

Clubs and scouts need to identify undervalued players and predict future performance while controlling wage bills and transfer costs. FootSight Analytics aims to answer questions like:

- Which players offer the best expected goals/assists per million euros of market value?
- Which young players have the highest gap between potential and current rating (high upside)?
- Which players are injury-prone and thus represent higher risk? Business objectives:

1. Improve transfer decision quality (reduce bad signings).
2. Prioritize scouting time toward high-ROI targets.
3. Forecast payroll inflation and budget impact.

4. Provide a shortlist (top N) for various tactical needs (e.g., creative midfielders under €20M).

3. Analytical Questions

1. Who are the top 50 undervalued players (sorted by goals/assists per million euros)?
2. Which players increased their overall rating most year-over-year (YoY)?
Which clubs sign players with the best long-term appreciation in market value?
3. What is the relationship between minutes played and player market value changes?
4. Which nationalities produce the highest average overall rating for attackers/midfielders/ defenders?
5. Which players have the highest injury-days per season and how does that affect minutes and market value?
6. Among players aged 18–23, who has the highest projected potential minus current rating?
7. What is the expected wage bill for the next season given current contracts and aging dynamics?
8. How do disciplinary issues (yellow/red cards) correlate with playing time or transfer value for defenders?
9. For each tactical profile (e.g., ball-playing defender), what are the top 10 candidates under €10M?
10. How do per-90 metrics (goals_per90, assists_per90) vary by position and age cohort?

4. KPIs

1. **Goals per 90 (G/90):** goals_per90.
2. **Assists per 90 (A/90):** assists_per90.
3. **Market Value (EUR, million):** market_value_million_eur.
4. **Value Efficiency:** (goals+assists) / market_value_million_eur.
5. **Overall Rating:** overall_rating (scouting/composite score).
6. **Potential Upside:** potential – overall_rating.
7. **Injury Risk:** injury_days per season.
8. **Salary Cost:** monthly_salary_k_eur (or annualized).
9. **Minutes Played:** minutes_played (availability).
10. **Disciplinary Index:** yellow_cards + 3*Red cards.